Natural logarithmic functions

1 Differentiate the following functions:

a $y = \ln x$

b $y = 7\ln x$

c $y = \ln 6x$

d $y = 2\ln 3x$

e $y = \ln 4x - \ln 3$

f $y = 4\ln 5x - x$

g $y = x^4 + 6\ln 7x$

h $y = 4\ln x - \dfrac{1}{x}$

i $y = 3\ln x + 5\ln 2x$

j $y = \ln(7x+3)$

k $y = \ln\left(x^4+2\right)$

l $y = \ln\left(x^2+7x+5\right)$

m $y = \ln\left(x^2-6x+9\right)$

n $y = \ln\left(2+4x+x^3\right)$

o $y = \ln\left(\left(4x^3+8x^2-9\right)^3\right)$

p $y = \ln\left(\dfrac{x}{7}\right)$

q $y = 4\ln\left(\dfrac{x}{5}\right)$

r $y = \ln\left(\dfrac{3}{x}\right)$

s $y = \ln(8-x) - 8$

t $y = \ln\left(t^2-5t\right)$

u $y = \ln\left(\dfrac{8}{x+4}\right)$

v $y = \ln\left(\dfrac{4}{3x+4}\right)$

2 Consider the function $y = \ln\left(\dfrac{x-3}{x+3}\right)$.

a Let $u = \dfrac{x-3}{x+3}$. Rewrite y in terms of u.

b Find $\dfrac{du}{dx}$.

c Find $\dfrac{dy}{du}$ in terms of x.

d Find $\dfrac{dy}{dx}$.

3 Consider the function $y = \ln x^2$, where $x > 0$.

a Rewrite the function without powers.

b Hence determine $y'(x)$.

c Find the value of x at which $y'(x) = \dfrac{1}{4}$.

4 Consider the function $y = \ln(5x-2)^4$.

a Rewrite the function without powers.

b Hence determine $y'(x)$.

c Find the exact value of x at which $y' = \dfrac{1}{3}$.

5 Differentiate the following functions:

a $y = \ln\sqrt{x}$

b $y = \ln\sqrt{7x}$

c $y = \ln\sqrt{2-5x}$

6 Consider the function $f(x) = \ln\left(\sqrt{x^2+1}\right.$ [redacted]

a Find $f'(x)$.

b Find $f'(2)$.

7 Consider the function $f(x) = 4\ln\left(4x^2+3\right)$.

a Find $f'(x)$.

b Find x, such that $f'(x) = 4$.

Multiple differentiation rules

8 Differentiate the following functions:

a $y = x\ln x$

b $y = (\ln x)^6$

c $y = \dfrac{\ln x}{x}$

d $y = \dfrac{x}{\ln x}$

e $y = x^3 \ln x^3$

f $y = 6x^4 \ln x$

g $y = x^5 \ln(x-5)$

h $y = (x+6)\ln(x+6)$

i $y = \left(x^2+6\right)\ln 2x$

j $y = e^x \ln x$

k $y = \ln e^{8x}$

l $y = e^{\ln x}$

m $y = e^{6x}\ln 4x$

n $y = \ln(\log_e 4x)$

o $y = \dfrac{\ln x}{e^x}$

p $y = \dfrac{\ln 3x}{5x^4+2}$

9 Consider the curve $y = x^3 \ln x$.

a Find the gradient function $\dfrac{dy}{dx}$.

b Find the exact value of the gradient at the point where $x = e^4$.

10 If $f(6) = 2$ and $f'(6) = 8$, find the value of $\dfrac{d}{dx}(\ln(f(x)))$ at $x = 6$.

11 Given that $f(x) = \ln(g(x))$, $g(2) = 4$ and $g'(2) = 9$, evaluate $f'(2)$.

12 Consider the function $f(x) = xe^{3x}$.

a Show that $xe^{3x} = e^{3x+\ln x}$.

b Hence, find $f'(x)$, without using the product rule.

13 Suppose that $g(x) = \dfrac{\ln x}{f(x)}$, for some func[redacted] (x).

- **a** Find an expression for $g'(x)$ in terms of $f(x)$ and its derivative $f'(x)$.
- **b** If $f\left(e^3\right) = 4e^3$ and $f'\left(e^3\right) = 2$, find the value of $g'(x)$ when $x = e^3$.

14 Consider the functions $f(x) = k\ln x$ and $g(x) = \ln kx$, where $k > 1$ is a constant.

- **a** Find $f'(x)$.
- **b** Find $g'(x)$.
- **c** State how many times faster $f(x)$ is increasing compared to $g(x)$.

15 Given the following expressions:

- $\ln x = x - 1 - \dfrac{(x-1)^2}{2} + \dfrac{(x-1)^3}{3} - \dfrac{(x-1)^4}{4} + \ldots$
- $\dfrac{1}{x} = 1 - (x-1) + (x-1)^2 - (x-1)^3 + (x-1)^4 - \ldots$

Prove that $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$.

Bases other than e

16 Consider the function $y = \log_3 x$.

- **a** Make x the subject.
- **b** Rewrite the expression for x with base e.
- **c** Find $\dfrac{dx}{dy}$.
- **d** Write $\dfrac{dx}{dy}$ in terms of x.
- **e** Hence find $\dfrac{dy}{dx}$.

17 Differentiate the following functions:

- **a** $f(x) = \log_2 x$
- **b** $f(x) = \log_2 4x$
- **c** $f(x) = \log_3 (2x - 4)$
- **d** $f(x) = \log_{10} (1 - x)$
- **e** $f(x) = \log_5 (4x + 2) - 6x$
- **f** $f(x) = \log_6 (2 - 7x)$